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Guideline No. 430: Diagnosis and management of preterm prelabour rupture of membranes

(En français : Directive clinique no 430 : Diagnostic et prise en charge de la rupture prématurée des membranes avant terme)

The English document is the original version. In the event of any discrepancy between the English and French content, the English version prevails.

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Weeks Gestation Notation: The authors follow the World Health Organization's notation on gestational age: the first day of the last menstrual period is day 0 (of week 0); therefore, days 0 to 6 correspond to completed week 0, days 7 to 13 correspond to completed week 1, etc.

Keywords: pregnancy complications; fetal membranes, premature rupture; premature birth; chorioamnionitis

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Informed consent: Everyone has the right and responsibility to make informed decisions about their care together with their health care providers. In order to facilitate this, the SOGC recommends that health care providers provide patients with information and support that is evidence-based, culturally appropriate, and personalized.

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KEY MESSAGES

1. Preterm prelabour rupture of membranes complicates approximately 3% of pregnancies; it causes about one-third of all preterm deliveries and is associated with high neonatal mortality and short- and long-term severe neonatal morbidity.
2. Prompt and accurate diagnosis of preterm prelabour rupture of membranes (based on patient history, physical examination, and conventional tests in addition to commercial tests in equivocal cases) should be considered for optimal maternal and fetal surveillance and management.
3. Because prematurity confers most of the fetal and neonatal risk in cases of preterm prelabour rupture of membranes, expectant management remains the most critical risk reduction strategy and represents the current standard of care in the absence of contraindications, such as infection, placental abruption, and cord accidents.

ABSTRACT

Objective: To provide clear and concise guidelines for the diagnosis and management of preterm prelabour rupture of membranes (PPROM)

Target Population: All patients with PPROM <37 weeks gestation

Benefits, Harms, and Costs: This guideline aims to provide the first Canadian general guideline on the management of preterm membrane rupture. It includes a comprehensive and up-to-date review of the evidence on the diagnosis, management, timing and method of delivery.

Evidence: The following search terms were entered into PubMed/Medline and Cochrane in 2021: preterm premature rupture of membranes, PPROM, chorioamnionitis, Nitrazine test, ferning, commercial tests, placental alpha microglobulin-1 (PAMG-1) test, insulin-like growth factor-binding protein-1 (IGFBP-1) test, ultrasonography, PPROM/antenatal corticosteroids, PPROM/Magnesium sulphate, PPROM/ antibiotic treatment, PPROM/tocolysis, PPROM/preterm labour, PPROM/Neonatal outcomes, PPROM/mortality, PPROM/outpatient/inpatient, PPROM/cerclage, previable PPROM.

Articles included were randomized controlled trials, meta-analyses, systematic reviews, guidelines, and observational studies. Additional publications were identified from the bibliographies of these articles. Only English-language articles were reviewed.

Validation Methods: The authors rated the quality of evidence and strength of recommendations using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. See Appendix A (Tables A1 for definitions and A2 for interpretations of strong and weak recommendations).

Intended Audience: All prenatal and perinatal health care providers.

SUMMARY STATEMENTS

1. Preterm prelabour rupture of membranes complicates approximately 3% of pregnancies and causes approximately one-third of all spontaneous preterm deliveries (*high*).
2. Preterm prelabour rupture of membranes is associated with high neonatal mortality and short- and long-term severe neonatal morbidity such as periventricular leukomalacia, bronchopulmonary dysplasia, necrotizing enterocolitis, retinopathy of prematurity, and adverse neurodevelopment outcomes (*high*).
3. The latency period from rupture of membranes to delivery is negatively correlated with gestational age at preterm prelabour rupture of membranes (*high*).
4. There is insufficient evidence on both the testing modality and the optimal frequency of testing to prevent adverse maternal and perinatal outcomes (*low*).
5. Bed rest is not beneficial in the setting of preterm prelabour rupture of membranes and has adverse maternal effects (*high*).
6. Serial monitoring of white cell count or other markers of inflammation have not been proven to be useful in the absence of other clinical signs of infection (*low*).
7. There is insufficient evidence to recommend hospital versus home management for preterm prelabour rupture of membranes (*low*).
8. Preterm prelabour rupture of membranes may complicate up to 38% of pregnancies in patients who have undergone cervical cerclage (*moderate*).
9. There are no randomized controlled trials comparing different management strategies (or timing) in the setting of previable preterm prelabour rupture of membranes (*low*).
10. Amniotic fluid volume at time of rupture may be helpful in counselling patients and families with previable preterm prelabour rupture of membranes, as anhydramnios and oligohydramnios are more frequently associated with pregnancy loss and pulmonary hypoplasia compared with normal amniotic fluid volumes (*low*).
11. There is conflicting evidence about the relationship between the gestational age at preterm prelabour rupture of membranes in a previous pregnancy and future pregnancy risks (*low*).
12. There is conflicting evidence about the effectiveness of prevention strategies for reducing complications in subsequent pregnancies for patients with a history of preterm prelabour rupture of membranes (*moderate*).
13. The reported recurrence risk of preterm prelabour rupture of membranes in future pregnancy ranges from 10% to 32%, but the most common complication in future pregnancy is preterm birth (34%−46%) (*moderate*).

RECOMMENDATIONS

1. The diagnosis of preterm prelabour rupture of membranes should be based on the combination of patient's history and physical examination by a sterile speculum with direct visualization of fluid in the posterior fornix (*strong, moderate*).
2. Digital exam should be avoided to reduce the risk of infection, unless the patient is in active labour (*strong, moderate*).
3. Multiple conventional tests (nitrazine, ferning test, and ultrasound evaluation of amniotic fluid volume) should be considered to confirm the diagnosis of preterm prelabour rupture of membranes when amniotic fluid is not visible at speculum examination (*strong, moderate*).
4. Commercial tests, particularly placental alpha microglobulin-1 (PAMG-1), should be considered following conventional tests in equivocal cases or used as the primary tests in rural and remote areas if other diagnostic options are not available or feasible (*strong, moderate*).
5. Once preterm prelabour rupture of membranes is diagnosed, the initial assessment should include maternal and fetal status with the principal purpose of ruling out active labour, infection (chorioamnionitis), placental abruption, or fetal distress, all conditions that warrant immediate delivery (*strong, high*).
6. Once preterm prelabour rupture of membranes is diagnosed, a vaginal/rectal swab should be obtained to test for group B *Streptococcus* colonization if not previously done within 5 weeks (*conditional, moderate*).

7. Expectant management with maternal and fetal monitoring should be offered to patients who have no contraindications to continuing the pregnancy (*strong, moderate*).
8. Given that the rate of preterm birth is highest immediately following preterm prelabour rupture of membranes, hospitalization is recommended for the first few days following diagnosis (*conditional, low*).
9. Based on the gestational age at preterm prelabour rupture of membranes and local capacity and resources, antenatal transfer to centres specialized in preterm care should be considered *(strong, high)*.
10. The optimal antibiotic regimen for preterm prelabour rupture of membranes remains unclear. If group B *Streptococcus* status is unknown or positive, the antibiotic regimen should include coverage for this pathogen (*strong, moderate*).
11. The following 2 antibiotic regimens may be used: 1) a macrolide (erythromycin, azithromycin, or clarithromycin) alone or associated with group B *Streptococcus* coverage for 2 days (if group B *Streptococcus* status is unknown or positive), or 2) a combination of ampicillin/amoxicillin and a macrolide independently of group B *Streptococcus* status. There are no data to support extending the antibiotic therapy beyond 10 days (*strong, moderate*).
12. Alternative antibiotic therapy can be considered based on local data on antibiotic resistance (*conditional, low*).
13. Antenatal corticosteroid therapy should be routinely administered to patients with preterm prelabour rupture of membranes at the time of diagnosis when gestational age criteria are met (*strong, moderate*).
14. There is insufficient evidence to support prolonged or recurrent use of tocolysis in the context of preterm prelabour rupture of membranes except for ensuring the full course of corticosteroids for 48 hours, or during transfer to a tertiary care centre in the absence of infection or abruption (*conditional, moderate*).
15. Magnesium sulphate administration for fetal neuroprotection is recommended following preterm prelabour rupture of membranes once the patient is in active labour or prior to indicated delivery, when gestational age criteria are met (*strong, moderate*).
16. If preterm prelabour rupture of membranes occurs before 34 weeks gestation, expectant management with careful monitoring is recommended at least until 35 weeks, in the absence of contraindications, such as infection, placental abruption, cord accident, or abnormal fetal health surveillance. There is conflicting evidence regarding the optimal timing of delivery in cases of preterm prelabour rupture of membranes during the late-preterm period (34^0 and 36^6 weeks gestation). If there is evidence of group B *Streptococcus* colonization, induction of labour should be considered (*conditional, moderate*).
17. In patients with preterm prelabour rupture of membranes and cervical cerclage, there is insufficient evidence on whether the cerclage should be removed or remain in situ. In the absence of signs of infection or other contraindications to retaining a cerclage, either option is reasonable (*conditional, low*).
18. Patients with previable preterm prelabour rupture of membranes should have a consultation with a maternal−fetal medicine specialist and neonatology specialist for comprehensive counselling about prognosis and risks, multidisciplinary management planning, and shared decision-making (*conditional, moderate*).

INTRODUCTION

Preterm prelabour rupture of membranes (PPROM) is defined as the spontaneous rupture of the fetal membranes before 37 weeks gestation and preceding the onset of labour.

PPROM complicates approximately 3% of pregnancies and causes approximately one-third of all preterm deliveries.[1] In Canada, preterm births account for 8% of all births, which translates into approximately 28 138 babies in 2020—an increase of almost 25% over the past decade.[2] PPROM is associated with high neonatal mortality and short- and long-term severe neonatal morbidity.[1,3–5]

The cause of membrane rupture is unknown, but recently, novel theories recognized that PPROM may result from complex and multifaceted pathways contributing to weakening of the membrane morphology through alteration of the collagen network and/or activation of matrix metalloproteinases triggered by bacterial products or pro-inflammatory cytokines.[6,7] Risk factors for PPROM are generally similar to those for spontaneous preterm labour with intact membranes, although microbial invasion of the amniotic cavity is identified, generally in a subclinical stage of infection, in up to one-half of all PPROM cases, particularly at earlier gestational ages[8–13] and in 70% of patients with PPROM who go into labour.[9]

PPROM is followed by a period of latency before the onset of labour, which can range from hours to several weeks.[3] The latency period is negatively correlated with gestational age at PPROM and is shorter in cases of oligohydramnios. Regardless of obstetric management or clinical presentation, 50% of patients with PPROM will deliver within 1 week.[14,15]

When PPROM occurs, in the absence of clinical signs of chorioamnionitis or labour, expectant management is recommended and generally consists of hospital admission with strict monitoring for signs of infection, placental abruption, fetal distress, and labour. Broad-spectrum antibiotics are routinely used in the management of PPROM to prolong pregnancy and to decrease maternal and neonatal mortality.[16,17] The management of PPROM represents a clinical challenge owing to the lack of specific signs that may predict the development of chorioamnionitis and latency for delivery.

Despite the high prevalence of preterm birth following PPROM, the optimal management of PPROM remains a topic of debate and is hindered by a lack of evidence. This guideline provides the first Canadian general recommendations on the management of preterm membrane rupture and includes a comprehensive view of the evidence and the latest research with respect to diagnosis, management, and timing of delivery.

DIAGNOSIS OF PPROM

PPROM is largely a clinical diagnosis, which should be based on a combination of the history, physical examination, and select laboratory markers (Table 1). The patient, in the absence of regular and painful contractions, may present with a "gush" of clear fluid from the vagina, continuous or intermittent leaking of fluid from the vagina, or the sensation of wetness at the level of perineum or vagina. A sterile speculum exam should be performed to identify spontaneous or provoked (after Valsalva or cough) pooling of amniotic fluid in the vaginal vault that is observed in about 60% of PPROM cases,[18] with a 12% false negative rate.[19] If PPROM is suspected, a digital exam should be avoided to reduce the risk of infection, unless the patient is suspected to be in active labour.[20,21] To improve the accuracy of PPROM diagnosis, conventional bedside confirmation tests have been introduced and performed at the time of speculum exam: the nitrazine test, which will turn positive based on the more alkaline pH amniotic fluid (7.0–7.3) compared with vaginal pH (3.8–4.2),[22] and arborization or ferning, which can be seen under the microscope.[23] The combination of maternal history, clinical examination, and conventional tests has shown an accuracy of 93%.[24] However, there can be an increased rate of false negative findings associated with prolonged rupture and increased false positive results secondary to vaginal contamination with blood, urine, microorganisms, or semen.[22–24] If bedside confirmatory testing fails to corroborate the clinical presentation, ultrasonography demonstrating oligohydramnios or anhydramnios may be helpful, but should be interpreted with caution as fluid leakage may be temporized by the presenting part, or be dynamically replaced by the fetus. If the diagnosis continues to be equivocal and the need for diagnosis outweighs the risks, ultrasound guided amniocentesis with indigo carmine dye infusion can be performed as the gold standard for the diagnosis of PPROM.[25] This procedure is invasive and carries the risk

ABBREVIATIONS

CRP	C-reactive protein
GBS	group B *Streptococcus*
PPROM	preterm prelabour rupture of the membranes
TVCL	transvaginal cervical length

Table 1. Performance of current available diagnostic tests for preterm prelabour rupture of membranes

Test	N	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	Accuracy (%)
Gold standard						
Intraamniotic dye injection	18	—	—	—	—	—
Clinical						
Clinical history	100	90	89	88	90	89
Conventional tests						
Nitrazine pH[125–129]	618	84-97	16-94	84-99	68-98	56-93
Arborization	642	51-96	71-100	61-100	73-91	63-96
Ultrasound						
AFI	151	94	91	—	—	92
Combined testing						
History + arborization + nitrazine	100	91	95	96	91	93
History + 2/3 of pooling, arborization, nitrazine	69	86	37	78	50	73
History + arborization + nitrazine + AFI	167	89-94	95-100	95-100	88-98	91-98
Commercial immunoassays						
PAMG-1[a128–133]	598	90-100	93-100	98-100	86-100	94-100
IGFBP-1[b]	513	90-100	98-99	97-98	90-100	93-99
Other vaginal fluid markers						
Prolactin (cut-off 20-30 μIU/mL)[134,135]	170	76-95	70-78	72-84	75-93	87
AFP (cut-off 35.9-125 μg/L)	180	94-100	94-100	94-100	94-100	—
β-hCG (cut-off 25-80 mIU/mL)	275	79-93	72-96	75-95	82-84	88
Lactate (cut-off 4.5 mmol/L)	200	86	92	92	87	—
Creatinine (cut-off 0.12-0.6 mg/dL)	339	89-100	90-100	89-100	89-100	—
Urea (cut-off 6.7-12 mg/dL)[136,137]	339	88-100	91-100	91-100	88-100	—
Haptoglobin (cut-off 94.5 mg/dL)	60	80	80	—	—	—
Thyroid hormone	60					
Total T4 (cut-off 0.866 μg/dL)		83	60	68	78	—
Free T4 (cut-off 0.079 ng/L)		90	70	75	88	—

[a]AmniSure ROM is the trade name of the commercial test for detecting PAMG-1.

[b]Actim PROM is the trade name of the commercial test for detecting IGFBP-1.

AFI: amniotic fluid index; AFP: α-fetoprotein; ß-hCG: beta subunit of human chorionic gonadotropin; IGFBP-1: insulin-like growth factor binding protein-1; PAMG-1: placental alpha microglobulin-1; T4: thyroxine.

of infection or rupture of the membranes and thus is not routinely performed.

In order to improve the diagnostic accuracy of PPROM and avoid undertreatment or overtreatment, several diagnostic tests have been developed in the last few decades.[24] Three systematic reviews evaluating novel methods of diagnosis of PPROM include non-conventional "commercial" immunoassay tests (for insulin-like growth factor-binding protein-1 [IGFBP-1] and placental alpha microglobulin-1 [PAMG-1]).[26–28] Compared with nitrazine or ferning test alone, both immunoassay tests showed increased accuracy but there was no statistical difference in equivocal cases.[26,27] The comparison between the 3 tests showed a higher accuracy for PAMG-1.[28] These systematic reviews should be considered with caution because the sample sizes of the included studies were small, the studies were all observational, and there was no reliable gold standard test against which the commercial tests were compared. A more recent prospective multicentre study compared the performance of PAMG-1 with the intra-amniotic infusion of indigo carmine dye in 140 patients with unknown membrane status. The results showed that the PAMG-1 test performed as well as the gold standard with a sensitivity of 100% (95% CI 0.95−0.99), specificity of 99.1% (95% CI 0.82−0.99), positive predictive value of 96.3% (95% CI 0.82−0.99), and negative predictive value of 100% (95% CI 0.97−1.0).[29]

Despite high costs, commercial tests should be considered as primary diagnostic tests in rural and remote where no other diagnostic options are available or feasible.

SUMMARY STATEMENTS 1, 2 AND RECOMMENDATIONS 1, 2, 3 and 4

INITIAL ASSESSMENT OF CONFIRMED PPROM

Once the diagnosis of PPROM is made, initial investigations should assess maternal and fetal wellbeing (Figure). This includes the assessment of maternal vitals (heart rate, blood pressure, temperature) and symptoms (contractions, vaginal bleeding, foul-smelling vaginal discharge, abdominal or uterine tenderness), maternal laboratory tests (white cell count, urine culture), and fetal assessment with ultrasound (fetal presentation, fetal biometry for gestational age, placental and cord location, evaluation of amniotic fluid volume, transvaginal cervical length measurement and biophysical profile if >28 weeks) and electronic fetal monitoring to monitor fetal status and uterine activity.[30,31]

A few studies have reported an association between a short cervical length (<2 cm) alone or in association with oligohydramnios and a shorter latency period for delivery.[32–34] The safety of transvaginal evaluation of cervical length in the context of PPROM has been consistently reported with no significant increase in endometritis, chorioamnionitis, or neonatal infection.[35,36] Specifically, 1 retrospective study of 171 patients with PPROM between 21^0 and 33^6 weeks gestation who underwent amniocentesis, found that a short cervix (<1.5 cm) was independently associated with an increased risk of intra-amniotic infection/inflammation and delivery within 7 days, which was independent from the presence of intra-amniotic inflammation.[34] Additionally, a prospective study that evaluated transvaginal cervical length and amniotic fluid index in 106 singleton pregnancies with PPROM between 23^5 and 33^6 weeks gestation found that a transvaginal cervical length >2 cm in association with amniotic fluid index >5 predicted a latency period of more than 7 days with 79% sensitivity and 83% specificity.[37]

At the diagnosis of PPROM, a vaginal/rectal swab to identify group B *Streptococcus* (GBS) colonization is recommended if not previously done within the last 5 weeks, although there is a lack of consensus on the optimal management of patients with PPROM who are GBS-positive.[38]

RECOMMENDATIONS 5 AND 6

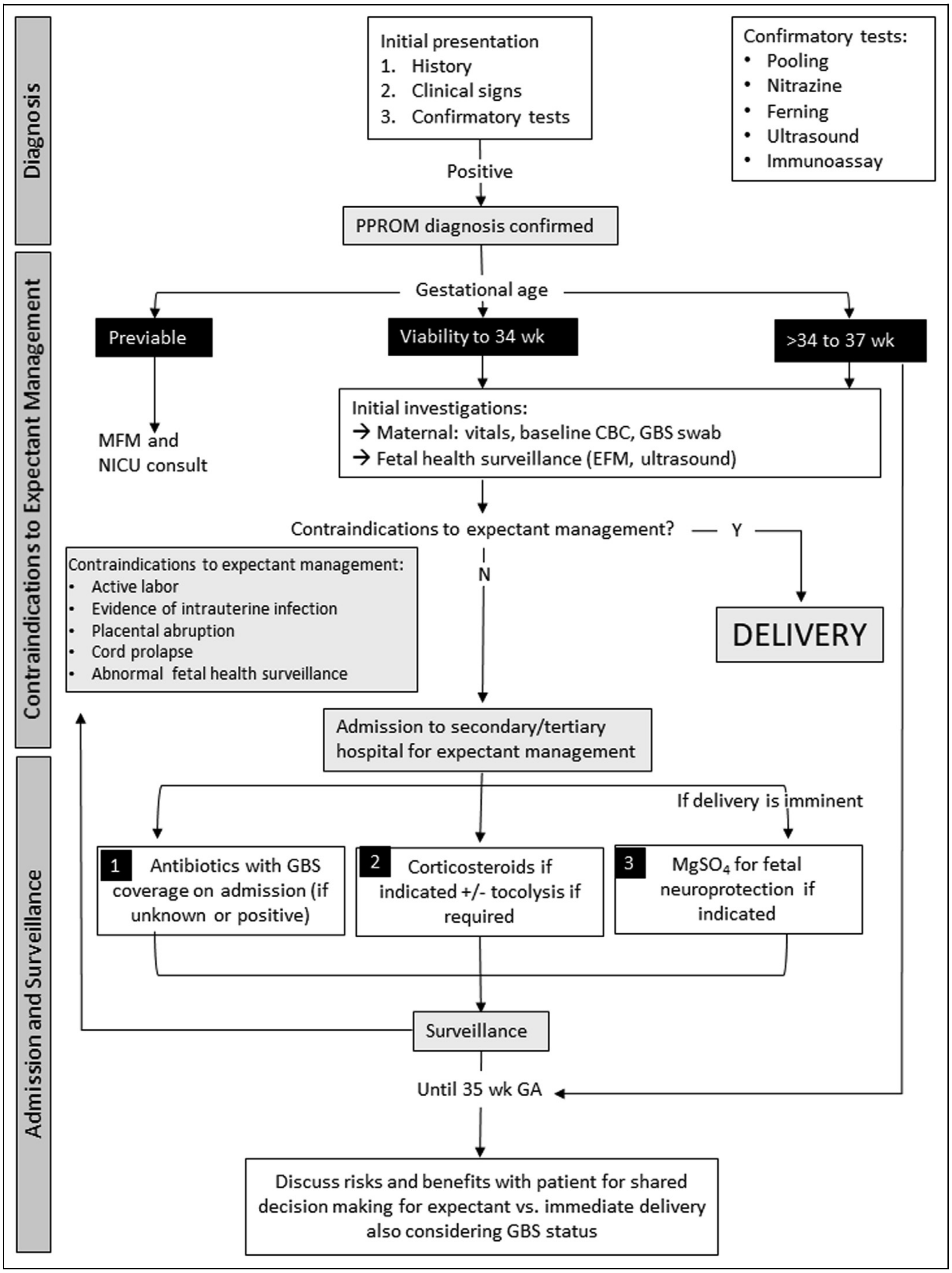
PPROM EXPECTANT MANAGEMENT

In the absence of maternal and fetal contraindications, expectant management should be offered to all patients before 37^0 weeks (Figure).[39] Expectant management improves neonatal survival by approximately 2% for each additional day of in utero maturation, with the optimal benefit between 24 and 27 weeks.[40] Expectant management currently involves inpatient monitoring for labour, signs of infection, placental abruption, cord prolapse, and fetal distress. However, there is no consensus regarding the optimal management strategy.

Fetal wellbeing is assessed with nonstress test and ultrasonography; however, there is scarce evidence on which to base the timing and frequency of these tests. A systematic review identified only 3 trials comparing fetal assessments in PPROM.[41] All 3 trials were conducted in the United States and employed different assessment methods, precluding the possibility of a meta-analysis. The largest of these trials involved 135 patients who underwent either a daily nonstress test (cardiotocography) or a daily biophysical profile ultrasound, and found that neither test had good sensitivity for predicting maternal or fetal infection nor adverse neonatal outcomes.[41] Oligohydramnios has been associated with an increased risk of perinatal infection, neonatal respiratory morbidity, and shorter latency period for delivery; however, when used alone, its predictive value is low.[42–45]

Maternal surveillance includes clinical assessment for signs of infection, including fever, maternal and fetal tachycardia, and foul-smelling vaginal discharge. There is no consensus on the definition of preterm intra-amniotic infection; generally, the diagnosis is inferred based on the same criteria used to define chorioamnionitis at term.[46] Serial monitoring of white cell count or other markers of inflammation have not been proven to be useful in the absence of other clinical signs of infection.[47–49] Moreover, white blood cell count should be interpreted with caution if measured between 24 hours and 3 days from corticosteroids administration.[50] A recent systematic review and meta-analysis on maternal inflammatory markers investigated the accuracy of maternal blood C-reactive protein (CRP), procalcitonin, and interleukin 6 (IL-6) in predicting chorioamnionitis in PPROM. Twenty-three observational prospective and retrospective studies, including a total of 1717 pregnancies complicated by PPROM, found that 902 had histological chorioamnionitis/funisitis. Sensitivity and specificity for CRP >20 mg/L was 59% (95% CI 48–69) and 83% (95% CI 74–89), respectively. Similar results

Figure. Proposed management of preterm prelabour rupture of membranes



CBC: complete blood count; EFM: electronic fetal monitoring; GA: gestational age; GBS: group B *Streptococcus*; MFM: maternal–fetal medicine; $MgSo_4$: magnesium sulfate; NICU: neonatal intensive care unit.

were found for CRP and the other markers considering all cut-offs.[47]

Bed rest has not been shown to be beneficial in the setting of preterm birth and has adverse maternal effects.[51] Two pilot trials have been recently published comparing bed rest versus mild activity in the setting of PPROM and found no differences in latency, chorioamnionitis, or neonatal outcomes.[51,52] However, no meaningful conclusions can be drawn from this research, as both studies were underpowered owing to low enrollment.

For the management of PPROM in the setting of active herpes simplex virus infection, refer to the SOGC Clinical

Practice Guideline No. 208, Guidelines for the Management of Herpes Simplex Virus in pregnancy.[53]

SUMMARY STATEMENT 3, 4, 5 AND 6 AND RECOMMENDATIONS 7, 8 AND 9

INPATIENT VERSUS OUTPATIENT CARE

Given the high rate of preterm birth following PPROM (about 30% within 48 h and about 50% within 7 d), it is reasonable to suggest hospitalization for the first few days following PPROM. Gestational age at PPROM should be considered because it is negatively correlated with latency for delivery.[14]

Currently, there is minimal evidence comparing hospital versus home management for PPROM. A Cochrane review published in 2014 identified only 2 small trials, 1 published in 1993 and the other consisting of conference abstracts from 1999. Only 116 patients were included, resulting in a sample size that was too small to reach adequate statistical power to detect meaningful differences between groups.[54]

Since then, there have been a handful of retrospective studies, mostly from France and Canada, comparing inpatient versus outpatient management.[55–57] Overall, these studies concluded that outpatient management was associated with a prolonged latency compared with inpatient management and represented an acceptable option given that both approaches had similar maternal and neonatal outcomes. However, there was considerable variability in the inclusion criteria, definition of outcomes, and protocols for inpatient versus outpatient management, which may have biased selection toward a lower-risk outpatient cohort. The largest retrospective study, conducted at 6 centres in France, included 587 patients with PPROM between 24^0 and 33^6 weeks who did not deliver for at least 48 hours. After propensity score matching, outpatient care was associated with similar rates of obstetrical and neonatal complications compared with inpatient management.[57]

SUMMARY STATEMENT 3, 7 AND RECOMMENDATION 8

PPROM and Antibiotics

There is clear evidence to support starting antibiotic therapy at admission for PPROM to increase the latency period for delivery from PPROM and to decrease maternal and neonatal morbidity.[58]

The mechanism of action of this treatment is thought to be not only direct antimicrobial activity but also nonspecific anti-inflammatory effect, especially for macrolides.[59] The choice of antibiotic therapy remains unclear because resistance patterns and practices (such as betamethasone injection, magnesium sulphate administration, and GBS screening) have evolved and, therefore, most prospective studies are outdated. The original randomized controlled trials (RCTs)[60,61] included a macrolide (erythromycin) with or without ampicillin and demonstrated a positive effect on neonatal morbidity and mortality but no difference at 7 years.[62] The use of amoxicillin-containing antibiotics in these trials was associated with prolongation of pregnancy, but also increased the risk of neonatal necrotizing enterocolitis.[60]

Bacteria isolated in cases of early onset neonatal sepsis in the context of PPROM are mainly *Escherichia coli*, *Staphylococcus,* and GBS.[63–67] However, molliculite infections, particularly *Ureaplasma* species, have been involved in the physiopathology of PPROM.[11,68,69] Molliculites are covered by macrolides, but this antibiotic class has a low transplacental transfer overall. Some experts have proposed oral azithromycin instead of erythromycin based on retrospective studies demonstrating similar latency and neonatal outcomes, but better tolerance and lower cost.[70–73] A 2021 meta-analysis compared the 2 macrolides in the context of PPROM <34 weeks and found a similar latency period and neonatal outcomes, but lower rate of clinical chorioamnionitis in the patients treated with azithromycin compared with those treated with erythromycin.[74]

A recent study demonstrated that treatment with clarithromycin in patients with PPROM and intra-amniotic infection diagnosed by amniocentesis was associated with a significant attenuation of the intra-amniotic inflammatory response at the follow-up amniocentesis.[59]

Table 2 shows the 2 different antibiotic regimens proposed for PPROM treatment, which are:

1) a macrolide (erythromycin, azithromycin, or clarithromycin) alone or associated with GBS coverage for 2 days (if GBS status is unknown or positive), or
2) a combination of ampicillin/amoxicillin and a macrolide independently of GBS status.

Data on the impact of expanding the antibiotic spectrum are limited. A trial that included 84 patients with PPROM comparing ampicillin plus roxithromycin to cefuroxime plus roxithromycin showed a significant longer latency period for delivery in the latter group, but with an unusually short latency period in the former (median 2.3 d).[75]

In cases of penicillin allergy, antibiotic therapy for GBS coverage should be chosen based on type of allergy and antibiotic susceptibility.[76]

Positive amniotic fluid cultures have been associated with a shorter latency period.[9,13] There are preliminary data showing that the adjustment of antibiotic therapy based on amniotic fluid culture could increase the latency period for delivery.[77] There are, however, no data to support the adjustment of antibiotic therapy based on maternal vaginal culture. Routine repetition of antibiotic prophylaxis is not recommended during the latency period.

RECOMMENDATIONS 10, 11 and 12

Table 2. Antibiotic regimens proposed for treatment of preterm prelabour rupture of membranes

Drug	Dosage, route, and duration
Macrolides	
Erythromycin OR	250 mg every 6 h for 10 d OR 250 mg IV every 6 h for 2 d, then 333 mg oral every 8 h for 5 d
Azithromycin OR	1 g oral single dose OR 500 mg to 1 g oral single dose, then 250 mg oral every 24 h for 4 d
Clarithromycin	500 mg oral every 12 h for 7 d
+ GBS coverage[a]	
Ampicillin AND	2 g IV every 6 h for 2 d
Amoxicillin OR	500 mg oral every 8 h for 5 d
Penicillin G	5 MU IV, then 2.5–3.0 MU every 4 h for 2 d

[a]Refer to SOGC Clinical Practice Guideline No. 298, The Prevention of Early-Onset Neonatal Group B Streptococcal Disease, for alternative treatment in case of penicillin allergy.

GBS: group B *Streptococcus*; IV: intravenous; MU: megaunit.

PPROM and Antenatal Corticosteroids

For dosage, timing, gestational age, and number of corticosteroid courses, refer to the SOGC Clinical Practice Guideline No. 364, Antenatal Corticosteroid Therapy for Improving Neonatal Outcomes.[78]

A single course of antenatal corticosteroid therapy at $24^{0/7}$–$34^{6/7}$ weeks gestation among patients at high risk for preterm birth has been shown to reduce perinatal mortality and morbidity with its benefits maximized when administered within 7 days of delivery.[78,79] However, another study found that administering antenatal corticosteroids at the time of PPROM resulted in more than half of patients receiving the medication outside the ideal timing and most of them receiving it too early.[80]

A meta-analysis of 17 trials with >1900 patients with PPROM confirmed that, when compared with a control group, the administration of corticosteroids is associated with significantly lower incidence of perinatal intraventricular hemorrhage and respiratory distress syndrome with no differences in necrotizing enterocolitis, neonatal sepsis, Apgar score <7 at 5 minutes, and chorioamnionitis between the steroid and control groups.[81]

RECOMMENDATION 13

PPROM and Tocolysis

Data regarding the use of tocolysis in cases of PPROM are controversial and reflect a wide variety of practice patterns.[82] There is insufficient evidence of any benefit from the use of tocolysis in the context of PPROM, with no demonstrated improvements in neonatal outcomes and conflicting results on prolongation of gestation, except in cases where the patient is receiving a full (48-h) course of corticosteroids or is being transferred to a tertiary care centre. A Cochrane review of 8 trials that included 408 cases of PPROM showed that, when compared with placebo, tocolysis was associated with longer latency period (mean difference 73 h; 95% CI 20–126; 3 trials, $n = 198$) and fewer births within 48 hours of PPROM (RR 0.55; 95% CI 0.32–0.95; 6 trials, $n = 354$), but with increased risks for Apgar score <7 at 5 minutes, need for ventilation support, and chorioamnionitis when PPROM occurred at <34 weeks gestation.[83]

Following the cited review, a single underpowered trial found that the prolonged use of tocolysis in PPROM without contractions was not associated with better perinatal outcomes and yielded no difference in latency duration when compared with placebo.[84] Furthermore, a secondary analysis of a national population-based prospective study in France looking at 803 cases of PPROM from 24 to 32 weeks gestation confirmed no prolonged latency period or improved obstetrical or neonatal outcomes associated with the use of tocolysis.[85]

Magnesium sulphate should not be used as tocolytic method. Two secondary analyses of RCTs of magnesium sulphate for prevention of cerebral palsy indicated, respectively, that magnesium sulphate does not appear to affect the latency period for delivery in patients with PPROM at <32 weeks not in labour[86] and is not associated with improvement in neurodevelopmental outcomes among children born prematurely and exposed to chorioamnionitis.[87]

RECOMMENDATION 14

PPROM and Magnesium Sulfate for Fetal Neuroprotection

For the use of magnesium sulphate in cases of PPROM, refer to the SOGC Clinical Practice Guideline No. 376, Magnesium Sulphate for Fetal Neuroprotection.[88] Patients at risk of imminent delivery should be given magnesium sulphate for fetal neuroprotection when gestational age criteria are met.[88]

Several trials have confirmed that the administration of magnesium sulphate is associated with a reduced rate of neonatal cerebral palsy and motor dysfunction in cases of preterm labour, or when preterm labour is planned or expected in the following 24 hours.[89] A recent systematic review of 6 trials that included 5917 patients found that magnesium sulphate infusion for patients at imminent risk for preterm birth reduced the rate of neonatal cerebral palsy compared with patients who did not receive magnesium sulphate (RR 0.68; 95% CI 0.54−0.85).[90]

RECOMMENDATION 15

PPROM and Timing of Delivery

Gestational age at the time of PPROM and at birth represent the primary predictors of short- and long-term neonatal outcomes, superseding the risk factor of intra-amniotic infection expected in cases where pregnancy is prolonged.[5,10,91]

A meta-analysis that included 12 trials, conducted between 1977 and 2016 (3617 pregnancies and 3628 babies), evaluated neonatal outcomes associated with planned early birth versus expectant management in pregnancies with PPROM between 25 and 37 weeks gestation.[92] Other than a lower incidence of chorioamnionitis in the early delivery group, the results indicated that the incidence of neonatal sepsis (RR 0.93; 95% CI 0.66−1.30; 12 trials, 3628 babies; evidence graded *moderate*) or proven neonatal infection confirmed by positive blood culture (RR 1.24; 95% CI 0.70−2.21; 7 trials, 2925 babies) was similar between the 2 approaches. In addition, early planned birth was associated with higher incidence of neonatal respiratory distress syndrome, need for ventilation, neonatal mortality, endometritis, admission to neonatal intensive care, and likelihood of cesarean delivery compared with expectant management. However, clinical suspicion of chorioamnionitis was reported in all studies and none of them confirmed the presence of intra-amniotic infection with amniocentesis or histological chorioamnionitis.

It should be noted that these studies did not include subgroup analysis based on gestational age at PPROM, so it is difficult to draw conclusions regarding the risks and benefits, particularly at lower gestational ages.[92]

The MICADO trial included a subgroup of PPROM occurring between 28 and 32 weeks gestation and found no difference in neonatal outcomes when comparing expectant management to early delivery.[93] However, the trial was interrupted because of difficulties with recruitment, and results were too underpowered to draw accurate conclusions.[93]

A meta-analysis of 3 RCTs (PPROMEXIL, PPROMEXIL-2, and PROMT) compared early delivery versus expectant management in cases with PPROM between $34^{0/6}$ and $36^{6/7}$ weeks gestation.[94] The trials included 2563 pregnancies and 2572 neonates, of which 1289 pregnancies (1291 neonates) were allocated to the immediate delivery group and 1274 pregnancies (1281 neonates), to the expectant management group.[95−97] No difference was found in the composite adverse neonatal outcome, which included probable or definitive neonatal sepsis, necrotizing enterocolitis, respiratory distress syndrome, stillbirth, and neonatal death (9.6% immediate delivery vs. 8.3% expectant management group; RR 1.20; 95% CI 0.94−1.55). Similarly, neonatal sepsis was similar between the 2 groups (2.6% immediate delivery vs. 3.5% expectant management; RR 0.74; 95% CI 0.47−1.15). Similar to the Cochrane review, immediate delivery was associated with higher incidence of cesarean delivery (RR 1.26; 95% CI 1.08−1.47), respiratory distress syndrome (RR 1.47; 95% CI 1.10−1.97), and neonatal intensive care unit or special care nursery admission (RR 1.17; 95% CI 1.11−1.23), but lower incidence of antepartum hemorrhage (RR 0.57; 95% CI 0.34−0.95) and chorioamnionitis (RR 0.21; 95% CI 0.13−0.35).[94] For late PPROM, while there appears to be sufficient evidence to support expectant management, 2 recent secondary analyses of the

PROMEXIL trial have highlighted some increased risks. One study addressed the neurodevelopmental outcomes of children at 2 years of age and found more neurodevelopmental problems after expectant management than after early induction of labour.[98] The other study suggested that patients with GBS colonization would benefit the most from immediate delivery owing to a significantly lower incidence of early onset neonatal sepsis seen with intervention (15.2% in expectant management vs. 1.8% in immediate delivery; RR 0.1 [95% CI 0.01–0.84]). In the subgroup of GBS-negative patients, the risk of early onset neonatal sepsis was lower and not significantly different between expectant management and immediate delivery.[99] These results are in contrast with those of a subgroup analysis of PPROMT, which did not find differences in early onset neonatal sepsis between GBS-positive patients and managed expectantly and those who delivered immediately (4% expectant management vs. 3% immediate delivery; RR 0.9 [95% CI 0.2–4.5]).[97] However, a recent systematic review, which included both PPROMT and PROMEXIL trials, confirmed the clinical and cost effectiveness of immediate delivery over expectant management in PPROM between 34 and 36 weeks for patients with GBS colonization.[100]

SUMMARY STATEMENT 3 AND RECOMMENDATION 16

PPROM in Patients with Cerclage

Although PPROM is a relatively frequent complication of cerclage, seen in as many as 38% of patients,[101] management of the cerclage after PPROM is controversial, with no prospective studies yet completed on this topic.

The only prospective, randomized, multicentre trial to date, comparing retention versus removal of the cerclage after PPROM between $22^{0/7}$ and $32^{6/7}$ weeks, was terminated at the second interim analysis because of concern regarding low power.[102] In their interim analysis, the authors found no differences in latency period for delivery, infection, or composite neonatal outcome, and a non–statistically significant lower incidence of complications related to infection was associated with immediate removal of cerclage.

There are a few retrospective cohort studies, mostly with small sample sizes, that found conflicting results regarding the safety and efficacy of retaining a cerclage after PPROM.[103–106]

Three literature reviews attempted to summarize available data on this topic with inconsistent conclusions. Giraldo-Isaza and Berghella suggested immediate cerclage removal as the preferred approach owing to an increased risk of maternal chorioamnionitis and neonatal mortality from sepsis seen with retention,[105] while Walsh et al. concluded that, in the absence of large, well designed studies, the decision to retain or remove cerclage in patients with PPROM should be done on an individualized based on maternal clinical status.[104] In a more recent meta-analysis, Pergialiotis et al. stated that the current evidence is insufficient to support the retention of cervical cerclage after PPROM.[103]

SUMMARY STATEMENT 8 AND RECOMMENDATION 17

PPROM at Previable Gestational Age

For information on the management of pregnancies at borderline viability, refer to the SOGC Clinical Practice Guideline No. 347, Obstetric management at borderline viability.[107]

PPROM at previable gestational age complicates approximately 0.1% of pregnancies and is associated with high fetal and neonatal morbidity and mortality, ranging from 46% to 95%.[108] Of note, iatrogenic previable PPROM after an invasive obstetric procedure such as amniocentesis or fetoscopy carries a better prognosis than spontaneous previable PPROM owing to a higher incidence of resealing of the amniotic sac, which has been reported to be as high as 72% in these cases.[7] To investigate the natural history of previable PPROM, Linehan et al. completed a single centre retrospective cohort study between 2007 and 2012, where termination of pregnancy in the absence of maternal compromise was unavailable. In their study, the mean gestation age at PPROM was 18 weeks, the average latency period for delivery was 13 days, and the mean gestational age at delivery was 20^5 weeks. Overall mortality was 95% (40/42), with only 10 infants born alive (23%; 10/42) and 2 surviving to discharge (5%).[108] A study conducted between 2012 and 2017 that included a cohort of 192 previable PPROM patients (mean gestational age at PPROM 20^5 weeks), who were expectantly managed with a mean delivery at 28.6 ± 5.1 weeks gestation, reported a neonatal survival rate to hospital discharge of 54% with a 10% risk of stillbirth.[109] Two Canadian studies have also described perinatal outcomes following previable PPROM. In a cohort of 99 previable PPROM pregnancies between 2009 and 2015, neonatal survival to hospital discharge was 27.5% in those choosing expectant management.[110] In another cohort of 104 cases between 2004 and 2014, 49.0% of neonates survived, of whom 23.3% had long-term complications.[111] PPROM after 22 weeks and a latency period of at least 7 days for delivery were the only 2 factors associated with survival without severe morbidity. Neonatal survival is also related to the residual amniotic fluid volume, with poorer outcomes and shorter latency associated with persistent severe oligohydramnios.[110,112] Pregnancy complications associated with previable PPROM include infection (chorioamnionitis, endometritis, sepsis), abruption, cord prolapse, intrauterine fetal demise, preterm birth, and need for cesarean delivery.[7] In addition to the usual sequelae of prematurity, previable PPROM is associated with pulmonary hypoplasia and musculoskeletal deformation.[113] The overall prevalence of pulmonary hypoplasia with previable PPROM is 30%,[114] which is inversely correlated with gestational age at PPROM and residual amniotic fluid volume.[115] The mortality rate for pulmonary hypoplasia is between 70% and 90%.[115] The incidence of musculoskeletal deformation/contractures is 7%–17%, which increases with prolonged oligohydramnios.[110,116] The diagnosis of PPROM is the same across gestations; however, a normal amniotic fluid volume by ultrasound does not rule out PPROM, as about 1 in 5 cases with confirmed previable PPROM have normal amniotic fluid volumes on ultrasound.[110] Once diagnosis is confirmed, the first step in patient management is counselling about the risks and benefits of expectant management versus pregnancy termination. Consultation with maternal–fetal medicine and neonatology specialists may assist with shared decision-making around pregnancy management. If the patient elects to proceed with expectant management and there are no contraindications, the next consideration is the timing of neonatal resuscitation. In general, outpatient management is reasonable after planning close maternal and fetal surveillance and providing detailed instructions on signs and symptoms of infection, labour, or placental abruption, which require prompt presentation to hospital. Admission to hospital, antibiotic prophylaxis, and antenatal corticosteroids are considered at viability or when neonatal intervention is requested. There are no available data on the risks and benefits of antibiotics at a gestational age of less than 24 weeks or on the regimen or timing of antibiotic prophylaxis (at diagnosis of PPROM vs. at reached viability).

Although a few observational studies found lower perinatal mortality associated with amnioinfusion, all RCTs published to date found no difference in perinatal mortality and neonatal morbidity, suggesting that amnioinfusion is not recommended.[117,118]

SUMMARY STATEMENTS 9, 10 AND RECOMMENDATION 18

Subsequent Pregnancy After PPROM

The influence of PPROM on future pregnancies is important to consider given current global efforts to prevent preterm births, of which about one-third of cases are caused by PROM.[119] In singleton pregnancies, recurrence of PPROM ranges from 10% to 32%.[120–122] There is also a 34%–46% increased risk of preterm birth in the subsequent pregnancy after PPROM. However, there is conflicting evidence about the relationship between gestational age at PPROM in the previous pregnancy and the risk or timing of complications in the subsequent pregnancy.[120–122] In one study, about 1 in 10 patients with a history of PPROM <27 weeks, experienced recurrent PPROM <27 weeks and 35% had a preterm birth in a subsequent pregnancy, while over half (59%) had no complications in their next pregnancy.[120] In that study, earlier gestational age at PPROM in the index pregnancy was one of the factors significantly associated with future preterm birth, along with older maternal age and negative vaginal culture for GBS.[120] In contrast, 2 other retrospective studies, 1 with 121 patients and 1 with 114 patients, found no relationship between the gestational age of membrane rupture in the index pregnancy and the risk of future PPROM and preterm birth.[121,122] Patients with prior previable PPROM had a 46% risk of preterm birth <37 weeks in subsequent pregnancy, including 23% of preterm births occurring <28 weeks and 17% occurring <24 weeks.[123] In this specific subgroup with prior previable PPROM, there were no differences in pregnancy outcomes between those who received cervical length surveillance or cerclage during the subsequent pregnancy and those who did not.[123] There is emerging literature to suggest that specific gene–environment interactions predispose certain patients to increased risk of PPROM and risk of recurrence. For instance, while bacterial vaginosis has inconsistently been associated with PPROM, one study identified that patients with bacterial vaginosis and a specific tumour necrosis factor-alpha (*TNF-α*) gene polymorphism are at a significantly increased risk of preterm birth following PPROM.[124] Further research is needed to fully evaluate the clinical utility of these gene–environment interactions in preventing PPROM and future pregnancy complications. Independent of the gestational age at which PPROM occurred, patients should be counselled to wait for at least 8 months before attempting a new pregnancy. There are no specific data available on the benefit of specific surveillance/intervention approaches, such as serial transvaginal ultrasound cervical length evaluation, progesterone supplementation, or cerclage in reducing recurrence of preterm birth in the subsequent pregnancy following PPROM. However, since a prior PPROM was included in all trials, it seems reasonable to offer serial ultrasound assessment of cervical length after 14 weeks and to consider progesterone supplementation and cerclage, if clinically indicated.

SUMMARY STATEMENTS 11, 12, AND 13

CONCLUSION

PPROM is a major cause of preterm birth and contributes to adverse short- and long-term perinatal outcomes. An accurate diagnosis of PPROM represents the first step toward safely and effectively managing this pregnancy complication. However, there is still insufficient evidence to suggest a preferable testing modality and the optimal frequency of testing to diagnose and monitor cases of PPROM, or the optimal management strategy and timing of delivery in PPROM cases. The proposed management strategy outlined in this guideline is based on the best available evidence and provides a useful tool for decision making surrounding expectant management versus immediate delivery, monitoring of complications during the latency period, and consideration of risk-reducing interventions before birth.

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APPENDIX A

Table A1. Key to Grading of Recommendations, Assessment, Development and Evaluation Quality of Evidence

Grade	Definition
Strength of recommendation	
Strong	High level of confidence that the desirable effects outweigh the undesirable effects (strong recommendation for) or the undesirable effects outweigh the desirable effects (strong recommendation against)
Conditional [a]	Desirable effects probably outweigh the undesirable effects (weak recommendation for) or the undesirable effects probably outweigh the desirable effects (weak recommendation against)
Quality of evidence	
High	High level of confidence that the true effect lies close to that of the estimate of the effect
Moderate	Moderate confidence in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Limited confidence in the effect estimate: The true effect may be substantially different from the estimate of the effect
Very low	Very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

[a]Do not interpret conditional recommendations to mean weak evidence or uncertainty of the recommendation.

Adapted from GRADE Handbook (2013), Table 5.1.

Table A2. Implications of Strong and Conditional recommendations, by guideline user

Perspective	Strong Recommendation • "We recommend that…" • "We recommend to not…"	Conditional (Weak) Recommendation • "We suggest…" • "We suggest to not…"
Authors	The net desirable effects of a course of action outweigh the effects of the alternative course of action.	It is less clear whether the net desirable consequences of a strategy outweigh the alternative strategy.
Patients	Most individuals in the situation would want the recommended course of action, while only a small proportion would not.	The majority of individuals in the situation would want the suggested course of action, but many would not.
Clinicians	Most individuals should receive the course of action. Adherence to this recommendation according to the guideline could be used as a quality criterion or performance indicator.	Recognize that patient choices will vary by individual and that clinicians must help patients arrive at a care decision consistent with the patient's values and preferences.
Policymakers	The recommendation can be adapted as policy in most settings.	The recommendation can serve as a starting point for debate with the involvement of many stakeholders.

Adapted from GRADE Handbook (2013), Table 6.1.